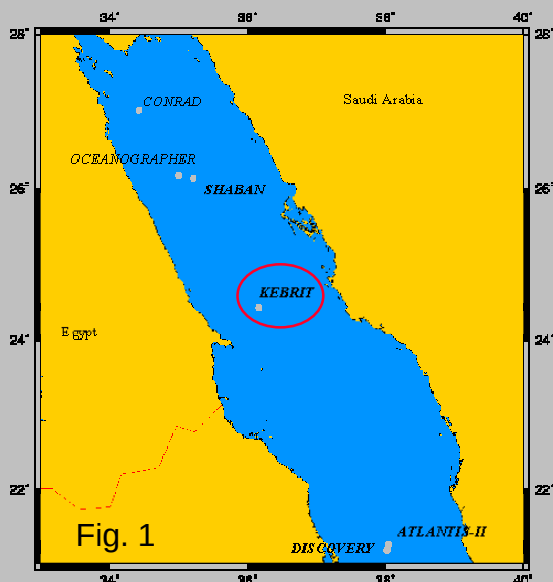


Thallium in massive sulfides from the Red Sea

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Bulk chemical XRF and INAA analyses of massive sulfides from the Kebrit Deep - one of the numerous brine filled depressions along the rift zone in the northern part of the Red Sea (Fig. 1: Location of the Kebrit Deep in the Red Sea) - yield unusually high Tl contents up to 3000 ppm. Nickel is also enriched in the samples (max. 150 ppm) and exhibits a nearly perfect correlation with Tl (corr. coeff. = 0,99). No obvious association with other elements such as Au, As and Sb was observed although this is often described from other low temperature epithermal systems.

Reflected light microscopic observations do not show any occurrences of discrete Tl-Ni-bearing phases in the Fe-, Zn- and Pb-sulfides of the Kebrit Deep (Fig. 2), suggesting that Tl and Ni are incorporated in at least one of the present ore minerals.

Detailed element mapping by LA-ICP-MS shows that galena and sphalerite are nearly free of Tl and Ni, whereas both elements are concentrated in some μm -wide growth zones of colloform-layered marcasite or adsorbed on its surface (Fig. 3). This implicates a pulsed input of a Tl and Ni enriched fluid into the brine filled Kebrit Deep during a limited time period.



Fig. 2: Colloform-layered Kebrit ore (marcasite) in reflected light with laser profiles. Length of profiles is 1,2 mm.

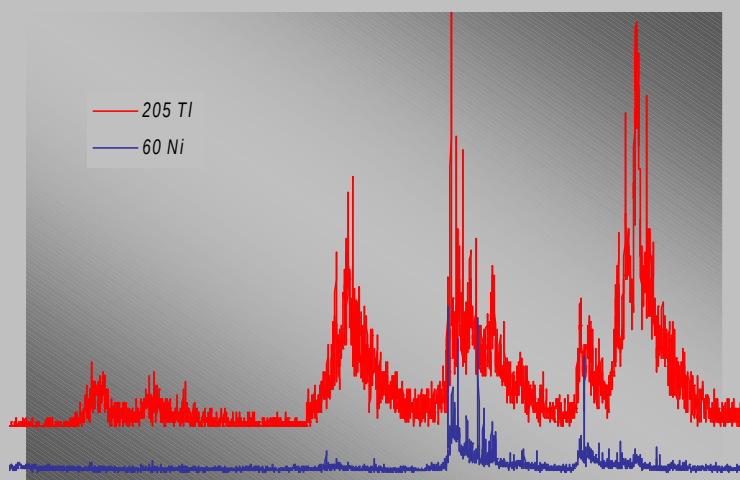


Fig. 3: LA-ICP-MS isotope profile of ^{205}Tl and ^{60}Ni on colloform marcasite shown in figure 2.